

• DETERMINISTIC EXECUTION
CORE

The Cost of Coordination, Engineered Away.

Animus Core is a C++17 zero-copy execution substrate for software that cannot tolerate jitter — replacing kernel-mediated queues and serialization boundaries with cache-aware, lock-free memory choreography between threads that share a machine.

30.3 ns

P50 ENQUEUE LATENCY

100 ns

TICK-TO-TRADE, P50 / P99

4.99×

CACHE ISOLATION GAIN

— THE PROBLEM

Every Message Your Stack Sends Today Passes Through a Tollbooth

Picture a highway engineered for uninterrupted, high-speed travel — then imagine every car forced through the same manned tollbooth, one at a time, regardless of how many lanes the highway actually has. Three tollbooths are hiding in most latency-sensitive systems — and a fourth lane that has none at all.



TOLLBOOTH 01 • OS MUTEX LOCKS

The Boom Barrier

A guard steps out, checks a ticket, and raises the gate — one car at a time — even when nobody else actually wants the road right now.

Technical: a kernel-arbitrated mutex/futex parks the thread and wakes it via a context switch — hundreds of ns to μ s per crossing, never the same twice.

COST: 100s ns- μ s • NON-DETERMINISTIC



TOLLBOOTH 02 • CACHE FALSE SHARING

The Single-Lane Bridge

Two cars in what should be independent lanes are funneled onto the same physical bridge deck — one driver braking makes the other brake too, though they were never meant to interact.

Technical: two unrelated variables sharing one 64-byte cache line force every core to invalidate and re-fetch the whole line on any write — silent contention, no lock in sight.

COST: SILENT THROUGHPUT LOSS • NO LOG



TOLLBOOTH 03 • PYTHON SERIALIZATION

The Border Customs Desk

Every vehicle is fully unpacked, inspected, stamped, and repacked before it may cross — even when nothing inside actually changed.

Technical: pickling, JSON encoding, and IPC copies pay a full serialize/deserialize tax at every process boundary — pure overhead, zero computation performed.

COST: FULL PICKLE/JSON TAX • EVERY MESSAGE



The Animus Express Lane

One dedicated lane, cache-isolated end to end — no barrier to stop for, no bridge deck to share, nothing to unpack.

30.3 ns • DETERMINISTIC • ZERO SYSCALLS

— THE ARCHITECTURE

How One Message Crosses the Ring — Three Steps

A single-producer/single-consumer ring buffer, walked step by step: the write, the hardware isolation that makes it safe, and the zero-copy read on the other side.

1

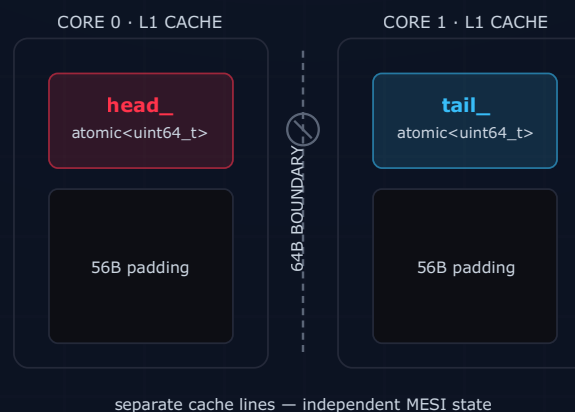
Ingest — Producer Write



The producer writes payload bytes directly into a pre-allocated slot, then publishes the new head index with a **release** memory barrier — guaranteeing the consumer never sees a torn write.

2

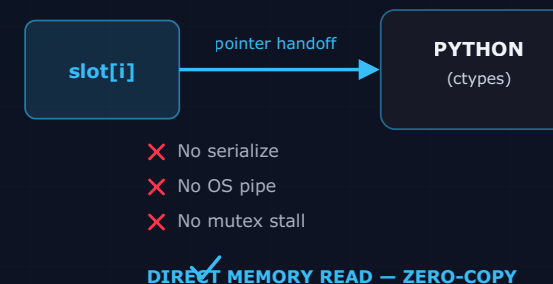
Cache Isolation — 64B Boundary



alignas(64) pins head_ and tail_ to their own cache lines. Core 0 writing head_ never invalidates Core 1's copy of tail_ — no ping-pong, no false sharing.

3

Zero-Copy Consumer — Into Python

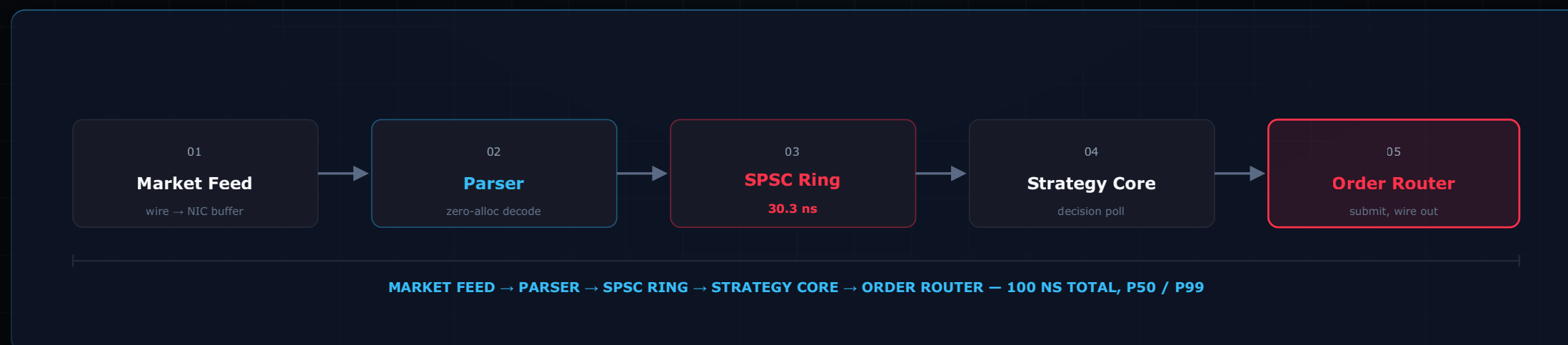


The consumer dereferences the same shared-memory pointer the producer wrote to — direct access, with **no kernel mediation**. Nothing is pickled, piped, or copied across the process boundary.

— THE ARCHITECTURE

The Tick-to-Trade Decision Pipeline

The same ring-buffer discipline, applied end to end: a market-data tick or sensor packet enters one side, a decision leaves the other, in a fixed, measured budget.

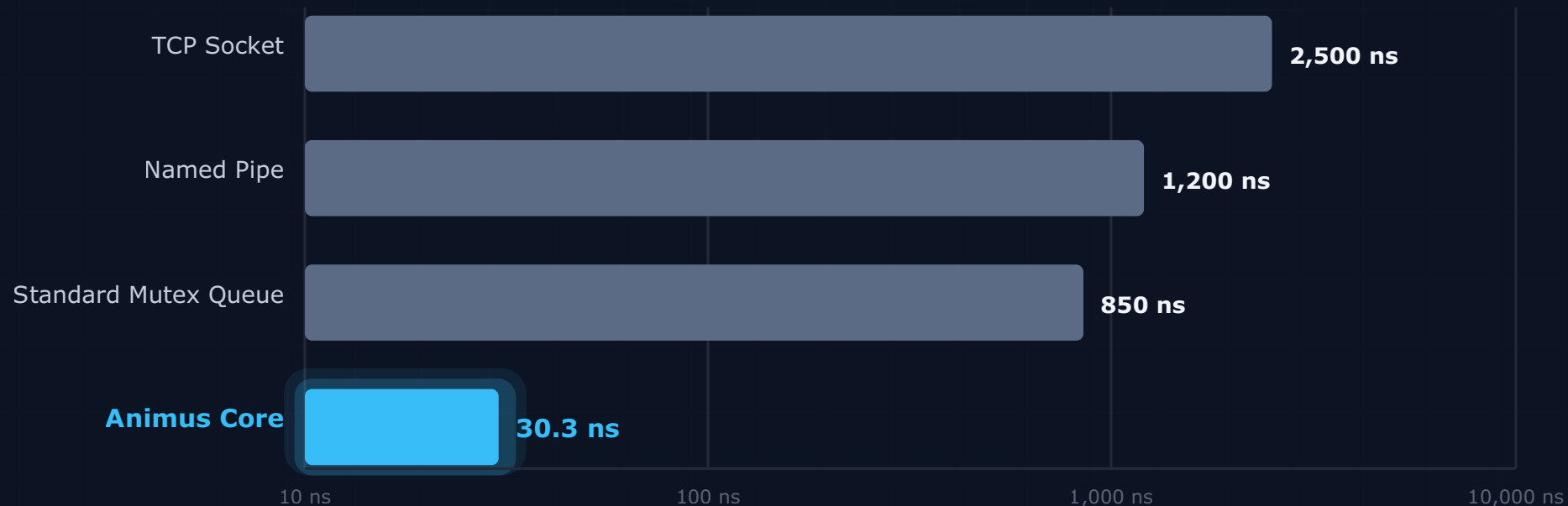


- 1 Feed handler and parser run **zero-allocation** — parsing happens in place, on the wire buffer.
- 2 The lock-free channel is the same cache-isolated SPSC ring from the previous page — no locks between ingest and strategy.
- 3 Strategy logic polls rather than blocks; execution submits the instant a decision is made — no scheduler in the loop.

— THE EVIDENCE

Four Orders of Magnitude, Not an Incremental Gain

Measured p50 latency to hand one message from a producer thread to a consumer thread — same host, same payload, same process — across four common transport mechanisms.



Baseline figures (TCP socket, named pipe, standard mutex queue) are representative order-of-magnitude values for common IPC/threading transports on modern commodity hardware, shown for scale — not a controlled head-to-head run on identical silicon. The Animus Core figure is founder-measured on development hardware; see the published benchmark datasheet for full methodology, host configuration, and raw sample counts.

VERIFIED PERFORMANCE

The Numbers Behind the Claim

Three measurements, taken directly from the compiled engine — not modeled, not projected.

30.3 ns

P50 Enqueue Latency

Single-message handoff through the SPSC ring buffer, warm cache, steady-state throughput.

100 ns

Tick-to-Trade, P50 / P99

Full round trip: market-data tick in → decision poll → execution submit out, 500,000-tick run.

4.99×

Cache Isolation Gain

Throughput improvement from alignas(64) padding versus an unpadding baseline, identical workload.

These are measured, reproducible results on the founder's own development hardware — not a third-party-audited or dedicated CPU-isolated benchmark rig. Reproduction scripts and raw output are included in the evaluation kit; institutional pilots are encouraged to re-measure on their own target hardware before committing to a design.

HARDWARE BASELINE: x86_64 Architecture • L1/L2 64B Cache Line Topology • Steady-State TSC Timing • Zero Dynamic Allocations during 500,000 continuous tick capture.

— WHERE DETERMINISM IS NOT OPTIONAL

Three Domains Where a Missed Tick Has a Real Cost

Animus Core targets systems where p99 tail latency — not average throughput — decides the outcome.



BUSINESS IMPACT • SLIPPAGE MITIGATION

Quantitative Finance Telemetry

Market-data feed handlers and order-routing telemetry where a single cache-line stall on the hot path is the difference between capturing a spread and reporting on one that already closed.



BUSINESS IMPACT • ZERO-JITTER CONTROL

Autonomous Robotics Control Loops

Sensor-to-actuator control loops that must complete a fixed-period cycle every time, not on average — jitter in the control loop is jitter in the physical motion of the machine.



BUSINESS IMPACT • MISSION ASSURANCE

Aerospace Sensor Fusion

Multi-sensor fusion pipelines (IMU, GPS, radar) merging readings from independent clocks into one coherent state estimate within a hard real-time budget, where a late tick is a safety event, not a support ticket.

— THE FOUNDER

Leadership & Engineering Vision



Alakshendra Roy

Founder & Chief Systems Architect
ANIMUS CORE SYSTEMS PRIVATE LIMITED

"I build deterministic C++ systems because most software's idea of 'fast' still budgets for a kernel to schedule it, a lock to grant it, or a serializer to translate it. **Animus Core exists to remove all three assumptions from the hot path** — letting cache coherence and memory ordering do the work an operating system used to do slower."

- Deterministic C++17/20 execution engines — lock-free, cache-aware, zero-copy by construction.
- Low-latency telemetry pipelines for market data, sensor fusion, and control-loop feedback.
- Cache coherence and false-sharing elimination treated as a first-class design constraint, not an afterthought.
- Building toward automated execution and telemetry infrastructure that behaves the same on the worst tick as the best one.

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GITHUB
github.com/alakshendraroy/AnimusCore_v1

— ENGAGEMENT MODEL

Licensed as a Precompiled Engine, Integrated Like a Library



Precompiled Binary Licensing

Animus Core ships as a compiled C-ABI dynamic library (.dll / .so). Integrating partners receive the binary and headers, not the source tree — integration is a linker-level exercise, not a build-system project.



Zero-Dependency Python SDK

A ctypes-based Python 3.8+ wrapper calls directly into the native engine's exported C-ABI — zero-copy buffer handoff, no serialization layer, no third-party Python packages to audit or pin.



Deployment Posture

Runs fully on-premises or air-gapped. No telemetry, license phone-home, or external network dependency is required for the core engine to operate.



Registered Activity Classification

Filed under MCA NIC codes **62011** and **62013** — Computer Programming, Software Licensing & Technical Support.

PLATFORM COMPATIBILITY

OS Linux (Enterprise RHEL / Ubuntu LTS)

OS Windows 10/11/Server

Toolchain C++17 / C++20 (C-ABI linkable)

Toolchain Python 3.8+ (ctypes)

Arch x86_64

Arch ARM64

PILOT TO PRODUCTION

1

Evaluation Binary & Mock Feeds



2

Shadow-Bus Benchmark



3

Direct In-Process Linking

Structured licensing — desk, enterprise, OEM-redistribution, and strategic tiers — is **scoped per engagement**.

[ASK FOR THE CURRENT TERM SHEET →](#)



Animus Core Systems

PRIVATE LIMITED · INDIA — INCORPORATION IN PROCESS

A deterministic, zero-copy execution substrate for the systems that cannot afford to wait for a lock, a copy, or a kernel to schedule them.

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This document is a confidential executive briefing prepared for evaluation purposes and does not constitute an offer or a binding commercial proposal. Benchmark figures are founder-measured and reproducible; see the accompanying benchmark datasheet for full methodology, host configuration, and raw sample counts.